

Evidence Based Medicine [EBM]

Dr. Irfan Adil Majid
Assistant Professor of Oral Medicine and Radiology

Lecture Learning Outcomes

- Define evidence-based dentistry [EBD]
- Describe different levels of evidence
- Outline the steps of practicing EBM

What is Evidence-Based Medicine/Dentistry [EBD]?

“Definition: "EBD is an approach to oral healthcare that requires the judicious integration of systematic assessments of clinically relevant scientific evidence, relating to the patient's oral and medical condition and history, with the dentist's clinical expertise and the patient's treatment needs and preferences.”

Traditional methods of gaining medical knowledge

- Continuing Medical Education (CME)
- Clinical Practice Guidelines (CPGs)
- Expert advice
- Reading
- **AI ???**

The Components of EBD

1. Best available evidence
2. Clinical experience and expertise of the clinician
3. Patient's needs and preferences

The 5 A's of Evidence-Based Dentistry

1. **A**sk: Formulate a clinical question
2. **A**cquire [obtain]: Search for the best evidence
3. **A**ppraise [Review]: Critically evaluate the evidence
4. **A**pply: Implement findings in clinical practice
5. **A**ssess: Evaluate the outcome

Ask: Formulating Clinical Questions-PICO

- P: Patient/Problem/Population
 - I: Intervention
 - C: Comparison
 - O: Outcome
 - Example: In patients with chronic periodontitis (P), how effective is scaling and root planning (I) compared to antimicrobial therapy (C) in reducing pocket depth (O)?
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Acquire [Obtain]: Searching for Evidence

- Databases: PubMed, Cochrane Library, EMBASE
- Types of studies (hierarchy of evidence):
 1. Systematic reviews and meta-analyses
 2. Randomized controlled trials (RCTs)
 3. Cohort studies
 4. Case-control studies
 5. Case series and case reports



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Appraising [Review] the Evidence

- Assess the validity, importance, and applicability of the evidence
- Critical appraisal tools:
 1. **CASP** [Critical Appraisal Skills Program] checklists]
 2. **GRADE** [Grading of Recommendations, Assessment, Development and Evaluation] system
- Consider potential biases and limitations of studies

Applying Evidence to Practice

- Integrate research evidence with:
 1. Clinical expertise
 2. Patient values and preferences
 3. Available resources
- Shared decision-making with patient

Barriers in Implementing EBD

- Time constraints
- Lack of skills in searching and appraising evidence
- Information overload
- Resistance to change in practice
- Limited access to resources

Benefits of EBD

- Improves quality of patient care
- Provides high standards of care
- Keeps practitioners up-to-date with current best practices
- Enhances clinical decision-making skills
- Promotes lifelong learning

EBD Resources for Saudi Dental Students

- Saudi Ministry of Health (MOH) Guidelines:
 1. Guidelines for infection control
 2. Saudi Clinical Preventive Guideline, and specific protocols for dental caries management (e.g., Silver Diamine Fluoride application)
- UpToDate
- EBD Literature and Journals
 1. Saudi Digital Library (SDL)
 2. King Saud University Dental Library
 3. ADA Center for Evidence-Based Dentistry
 4. Cochrane Oral Health Group
 5. Journal of Evidence-Based Dental Practice
 6. Saudi Dental Journal
- Remember: EBD is about providing personalized dental care based on the most current scientific knowledge while considering patient preferences and clinical expertise

Thank you